

Module 4: Data Modeling & DAX with Power Pivot

1. Data Modeling

Data Modeling is the process of organizing multiple related tables into a structured model for analysis.

Imagine you have customer data in one notebook, product data in another, and sales data in a third. Instead of copying everything into one huge table, you connect them using common IDs so they work together.

Example

- Customers Table
- Orders Table
- Products Table

Connect them using:

Customer ID and Product ID

Relationships Between Tables

A relationship is a connection between two tables using a common column (key).

Think of relationships like family connections. A child knows their parents because of a relationship. Similarly, Excel knows which customer placed which order because of a common ID.

Example

Customers

Customer ID	Name
-------------	------

101	Ali
-----	-----

Orders

Customer ID	Order
-------------	-------

Customer ID Order

101 Laptop

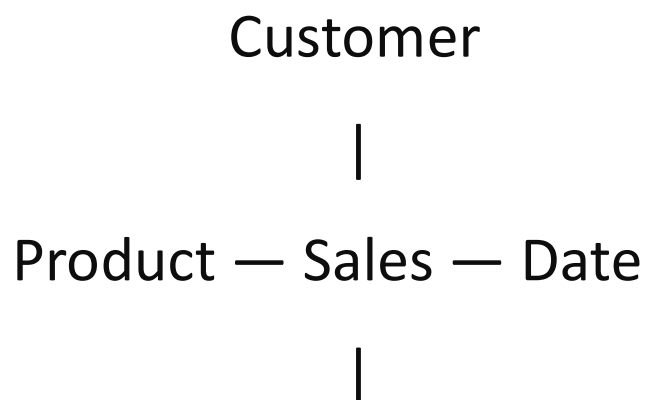
Customer ID links both tables.

Star Schema

A Star Schema is a data model where one central fact table connects directly to multiple dimension tables.

Imagine the sun in the center and planets around it. The Sales table is the center, while Customer, Product, and Date tables surround it.

Example



Employee

Snowflake Schema

A Snowflake Schema is an extension of the Star Schema where dimension tables are further divided into related tables.

Instead of one simple branch, the information spreads into more branches, like a snowflake.

Example

Sales

|

Customer

|

City

|

Country

2. Create Relationships

Creating Relationships means linking tables through common columns.

Instead of repeating customer names in every sales record, Excel remembers the connection through Customer ID.

Example

Sales.CustomerID



Customers.CustomerID

3. KPIs & Hierarchies

KPI (Key Performance Indicator)

A KPI is a measurable value used to evaluate business performance.

A KPI tells us whether the business is doing well or not.

Example

- . Total Sales
 - . Profit
 - . Revenue
 - . Customer Satisfaction
-

Hierarchy

A Hierarchy organizes data into different levels.
Like folders inside folders.

Example

Year



Quarter



Month



Day

4. DAX (Data Analysis Expressions)

DAX is a formula language used in Power Pivot and Power BI to perform calculations and create business logic.

If Excel formulas are calculators, DAX is a smart calculator that works across multiple connected tables.



Example

Calculate Total Profit

Average Sales

Running Total

Year-over-Year Growth

5. Calculated Columns vs Measures

Calculated Column

A Calculated Column creates a new value for every row.

Each row gets its own permanent calculation.

Example

$\text{Profit} = \text{Sales} - \text{Cost}$

Every row has its own Profit value.

Measure

A Measure performs calculations only when needed in reports.

Instead of storing values for every row, it calculates the result dynamically.

Example

Total Sales

Average Profit

6. Aggregation Functions

Aggregation functions summarize multiple values into one result.

Instead of looking at 10,000 numbers individually, aggregation gives one meaningful summary.

Example

- . SUM
- . AVERAGE
- . COUNT
- . MIN
- . MAX

7. Logical Functions

Logical functions make decisions based on conditions.

Just like humans decide "If this happens, do that," DAX can make decisions too.

Example

IF(Sales > 1000,"High","Low")

8. Filter Functions

Filter functions return only the data that matches specific conditions.

Imagine searching for only "Laptop" sales from thousands of products.

Example

Show only:

Region = Karachi

9. Relationship Functions

Relationship functions retrieve related data from connected tables.

They allow one table to borrow information from another linked table.

Example

Sales table retrieves Customer Name from Customers table.

10. Time Intelligence Functions

Time Intelligence functions analyze data across dates and time periods.

Instead of just showing today's sales, they compare today with last month, last year, or calculate year-to-date performance.

Example

- Year-to-Date Sales
 - Last Month Sales
 - Same Month Last Year
-

11. Math & Statistical Functions

These functions perform mathematical and statistical calculations.

They help analyze numbers instead of just displaying them.

Example

- Average
- Standard Deviation

- Variance
 - Square Root
-

12. Text Functions

Text functions manipulate or modify text values.

They clean, combine, or extract information from text.



Example

UPPER()

LOWER()

LEFT()

RIGHT()

CONCATENATE()

13. Important DAX Functions

CALCULATE()

Changes the context of a calculation before returning the result.

Think of CALCULATE as saying, "Calculate this, but only under these conditions."

Example

Calculate sales only for the **North Region**.

FILTER()

Returns rows that satisfy a condition.

Like applying a smart filter before calculating.

Example

Sales greater than **1000**.

RELATED()

Retrieves a value from a related table.

It's like asking your neighboring table for information because you're connected.

Example

Sales table gets Product Name from Product table.

SUMX()

Calculates an expression for each row and then sums the results.

Instead of simply adding numbers, it performs a calculation on every row first, then adds everything together.

Example

Profit = Quantity × Price

for every row, then calculates the total profit.

DATESYTD()

Returns dates from the beginning of the year to the current date.

Shows everything from January 1st until today.

Example

Year-to-Date Sales.

SAMEPERIODLASTYEAR()

Returns the same date range from the previous year.

Perfect for comparing this year's performance with the same period last year.

Example

Sales in **July 2026** compared with **July 2025**.

14. Visual-Level Filtering

Applies filters to only one visual without affecting others.

Imagine a dashboard with five charts. You want only one chart to show **Karachi Sales**, while the other four remain unchanged.

Example

Bar Chart → Karachi only

Pie Chart → All Pakistan

15. Performance Best Practices

Performance Best Practices are techniques used to make data models and reports faster and more efficient.

Instead of making your report heavy and slow, you organize data and write efficient calculations so dashboards open quickly.

Example

- Remove unnecessary columns.
- Use Star Schema instead of overly complex models when appropriate.
- Avoid unnecessary calculated columns.
- Prefer efficient DAX measures.
- Keep the data model as simple as possible.